

Global Climate Events Economic Impact

Project Documentation

Course: Data Analyst Specialist Track - DEPI

Dataset Period: 2020–2025

Dataset Source: Kaggle — Global Climate Events

Team Members

Mohamed Emad	Team Leader — SQL & Python Analysis
Ahmed Bahaa	SQL Analysis, Python Analysis
Shaimaa Abdelrahman	Excel Dashboard, Power BI, Tableau
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1 Project Planning & Management

1.1 Project Proposal

Project Overview

The **Global Climate Events Economic Impact** project focuses on the systematic analysis of worldwide climate disasters and their associated economic and human consequences between 2020 and 2025. The project transforms raw climate-event data into actionable insights through a rigorous pipeline encompassing data cleaning, exploratory analysis, KPI generation, statistical trend analysis, and multi-tool dashboard development.

The project was implemented independently using five analytical environments — Microsoft Excel, SQL, Python, Power BI, and Tableau — each executing the complete analysis workflow from raw data ingestion through to interactive dashboard delivery.

1.2 Project Objectives

- Consolidate and clean global climate event records spanning 2020–2025 into a reliable, analysis-ready dataset.
- Quantify the human toll of climate disasters — total deaths, injuries, and affected populations — disaggregated by event type, season, and geography.
- Measure the economic burden of climate events, including direct losses, infrastructure damage scores, and international aid coverage.
- Identify temporal and seasonal patterns that determine peak risk periods and the event types most strongly associated with elevated severity.
- Evaluate emergency response efficiency and its relationship to casualty outcomes.
- Deliver interactive, stakeholder-ready dashboards using Power BI, Tableau, and Excel that support data-driven disaster preparedness and policy decisions.
- Produce standardised, well-documented code and analytical artefacts that can be replicated or extended in future research.

Project Scope

In Scope

- Data cleaning and preprocessing across all five tools.
- Exploratory data analysis (EDA) and summary statistics.
- KPI generation: Total Events, Total Deaths, Total Economic Impact, Average Severity, Average Response Time.
- Statistical and trend analysis (seasonal, geographic, temporal).
- Dashboard development with interactive filters.
- Visualization and reporting.

Out of Scope

- Machine learning predictive models.
- Real-time data streaming or live API integration.

- Web or mobile application deployment.

1.3 Project Plan & Timeline

The project was executed over a structured six-phase timeline. Each phase had defined deliverables and was reviewed by the full team before progressing to the next stage.

Phase	Activity	Duration	Deliverable
1	Project Scoping & Dataset Acquisition	Week 1	Project proposal, Kaggle dataset downloaded
2	Data Cleaning & Preprocessing	Week 2	Cleaned dataset (all 5 tools), validation report
3	Exploratory Data Analysis	Week 3	EDA notebooks, summary statistics, KPI definitions
4	Statistical & Trend Analysis	Weeks 4–5	SQL queries, Python analysis scripts, trend charts
5	Dashboard Development	Weeks 5–6	Power BI, Tableau & Excel interactive dashboards
6	Testing, Documentation & Presentation	Week 7	QA report, project documentation, final presentation

1.4 Task Assignment & Roles

Team Member	Role	Responsibilities
Ahmed Bahaa	Data Engineer	SQL analysis, Python data analysis & visualisation
Mohamed Emad	Team Leader	SQL analysis, Python analysis, team coordination
Shaimaa Abdelrahman	BI Developer	Excel dashboard, Power BI, Tableau
Amal Ayman	BI Developer	Excel dashboard, Power BI, Tableau
Basant Abdelhakim	BI Developer	Excel dashboard, Power BI, Tableau

1.5 Risk Assessment & Mitigation Plan

Risk	Likelihood	Impact	Mitigation Strategy
Missing values in dataset	High	Inaccurate analysis	Data cleaning, imputation, and validation checks
Incorrect data types	Medium	Calculation errors	Data preprocessing and explicit type conversion
Dashboard performance issues	Low	Slow visualisation	Optimise DAX/calculated fields; reduce dataset size
Team coordination issues	Medium	Delayed progress	Task distribution via shared tracker; weekly sync
Data inconsistency across tools	Medium	Different outputs	Standardised cleaning script applied before each tool

1.6 Key Performance Indicators (KPIs)

KPI	Definition	Business Question Answered
Total Events	Count of all recorded climate events	How frequently do climate disasters occur?

KPI	Definition	Business Question Answered
Total Deaths	Sum of fatalities across all events	What is the human mortality cost of climate disasters?
Total Economic Impact (M USD)	Aggregate economic loss in millions USD	What is the financial scale of climate disasters?
Average Severity (1–10)	Mean severity score across events	How intense are climate events on average?
Average Response Time (hrs)	Mean emergency response time in hours	How quickly are disaster responses mobilised?
Aid Coverage (%)	International aid as % of economic loss	Is international support proportional to losses?

2 Requirements Gathering

2.1 Stakeholder Analysis

Stakeholder	Primary Needs	Dashboard Focus
Analysts	Accurate, granular climate insights	Raw KPIs, data tables, trend lines
Decision Makers	High-level visual dashboards and KPIs	Executive summary page, geo maps
Researchers	Trend and impact analysis over time	Time-series charts, seasonal breakdowns
Emergency Management Teams	Risk assessment, response evaluation	Response time charts, severity heat maps

2.2 User Stories & Use Cases

Representative User Stories

- *As a decision maker*, I want to analyse climate disaster trends so that I can improve disaster preparedness strategies.
- *As an analyst*, I want to filter events by country and event type so that I can compare regional vulnerability.
- *As a researcher*, I want to view seasonal economic impact trends so that I can identify peak-risk periods.
- *As an emergency manager*, I want to correlate response time with casualty counts so that I can evaluate intervention effectiveness.

Use Cases

- View total climate events by year, season, and event type.
- Analyse economic impact by country and continent.
- Compare fatality and injury rates across disaster categories.
- Investigate seasonal patterns in event frequency and severity.
- Evaluate emergency response efficiency relative to human impact.
- Assess international aid coverage versus actual economic losses.

2.3 Functional Requirements

- Import and clean the raw Kaggle climate dataset in all five tools.
- Generate KPIs: Total Events, Total Deaths, Total Economic Impact, Avg Severity, Avg Response Time.
- Perform aggregations by event type, country, continent, year, and season.
- Create interactive dashboards with drill-down filters.
- Visualise trends using line charts, bar charts, heat maps, and geo maps.
- Validate outputs for consistency across all five analytical environments.

2.4 Non-Functional Requirements

Requirement	Description	Target
Performance	Dashboard loading speed	< 3 seconds per page
Reliability	Calculation accuracy	Zero tolerance for formula errors
Usability	Dashboard navigation	Intuitive for non-technical stakeholders
Scalability	Dataset size handling	Supports datasets up to 100k rows
Maintainability	Code readability	PEP-8 compliant Python; named SQL CTEs

3 System Analysis & Design

3.1 Problem Statement

Climate disasters are increasing in both frequency and intensity globally, inflicting significant economic losses and human casualties. The sheer volume and multi-dimensional nature of climate event data makes manual analysis impractical and prone to error. Without systematic analysis, it is difficult for decision-makers and emergency management teams to identify risk patterns, allocate resources efficiently, or measure the effectiveness of disaster-response mechanisms.

3.2 Project Goals

- Simplify and automate climate data analysis using industry-standard tools.
- Provide meaningful, evidence-based insights into the economic and human impact of climate events.
- Support data-driven decision making through interactive dashboards and clear KPIs.
- Visualise spatial, temporal, and categorical patterns in climate disaster data.
- Establish a reusable analytical framework applicable to future climate datasets.

3.3 Software Architecture

The project follows a layered **data analytics workflow architecture**, implemented independently within each of the five analytical tools:

1 — Data Collection

Raw climate event dataset sourced from Kaggle and loaded into each tool environment (Excel workbook, SQL database, Python DataFrame, Power BI model, Tableau data source).

2 — Data Cleaning

Missing value treatment, duplicate removal, data type correction, outlier identification, and column standardisation applied uniformly across all tools.

3 — Data Analysis

Aggregations, KPI calculations, group-by operations, seasonal decomposition, correlation analysis, and response-time evaluation.

4 — Visualisation & Dashboarding

Interactive dashboards with filters by country, event type, severity level, response time, and infrastructure damage. Includes maps, trend lines, bar charts, heat maps, and KPI cards.

3.4 Technology Stack

Technology	Primary Usage	Key Outputs
Microsoft Excel	Data cleaning, pivot analysis, dashboarding	Pivot tables, KPI dashboard, charts
SQL	Querying, aggregations, schema design	Star-schema DB, analytical query scripts
Python (Pandas, Matplotlib, Seaborn)	EDA, statistics, visualisation	Jupyter notebooks, multi-page dashboard

Technology	Primary Usage	Key Outputs
Power BI	Interactive dashboard creation, DAX measures	Multi-page interactive report (.pbix)
Tableau	Data visualisation and story-telling dashboards	Tableau workbook with calculated fields

3.5 Database Design & Data Modelling

Star Schema Design

The SQL database was designed as a **star schema**, with a central fact table surrounded by five dimension tables. This design optimises query performance for aggregation-heavy analytical workloads and ensures clean separation of measurable facts from descriptive attributes.

Fact Table — `fact_global_climate_events`

The fact table is the core of the schema, storing one row per climate event. It holds all quantitative measures and foreign keys linking to dimension tables:

Column	Data Type	Description
event_id	VARCHAR(10)	Primary key — unique event identifier
date_id	INT	FK to dim_date
location_id	INT	FK to dim_location
event_type_id	INT	FK to dim_event_type
infrastructure_damage_id	INT	FK to dim_infrastructure_damage
response_time_id	INT	FK to dim_response_time
severity_category_id	INT	FK to dim_severity
severity	INT	Severity score (1–10)
duration_days	INT	Event duration in days
affected_population	INT	Total population affected
deaths	INT	Number of fatalities
injuries	INT	Number of injuries
economic_impact_million_usd	DECIMAL(6,2)	Economic loss in millions USD
infrastructure_damage_score	DECIMAL(6,2)	Infrastructure damage score
response_time_hours	INT	Emergency response time in hours
international_aid_million_usd	DECIMAL(6,2)	Aid received in millions USD
total_casualties	INT	Deaths + injuries combined
impact_per_capita	DECIMAL(4,2)	Economic impact per capita
aid_percentage	DECIMAL(4,2)	Aid as % of economic loss

Dimension Tables

dim_date

Purpose: Stores full date hierarchy enabling year, quarter, month, and season filtering.

- date_id INT (PK)
- full_date DATE
- year INT
- quarter INT
- month INT
- month_name VARCHAR(20)
- day INT
- day_of_week INT
- day_name VARCHAR(20)
- week_of_year INT
- season VARCHAR(10)
- is_weekend TINYINT(1)

dim_location

Purpose: Geographic hierarchy supporting country and continent-level aggregations and geo-maps.

- location_id INT (PK)
- country VARCHAR(100)
- continent VARCHAR(50)
- latitude DECIMAL(10,6)
- longitude DECIMAL(10,6)

dim_event_type

Purpose: Classifies events (e.g., Earthquake, Flood, Wildfire, Hurricane, Heatwave, Drought, etc.).

- event_type_id INT (PK)
- event_type VARCHAR(100)

dim_severity

Purpose: Maps numeric severity scores to categorical levels (Low, Moderate, High, Catastrophic).

- severity_category_id INT (PK)
- severity INT
- severity_category VARCHAR(20)

dim_response_time

Purpose: Classifies response speed (Very Fast, Fast, Normal, Slow) for response-efficiency analysis.

- response_time_id INT (PK)
- response_time_hours INT
- response_time_category VARCHAR(30)

dim_infrastructure_damage

Purpose: Categorises structural damage (Light, Moderate, Strong) per event.

- infrastructure_damage_id INT (PK)
- infrastructure_damage_score DECIMAL(6,2)
- infrastructure_damage_level VARCHAR(30)

4 Implementation

4.1 Source Code Organisation

The project codebase was structured for clarity, reproducibility, and maintainability. Both the SQL scripts and Python notebooks follow consistent naming conventions and are organised into logical, self-contained sections.

- **Data Cleaning:** Missing value imputation, type conversion, duplicate removal, column renaming.
- **Exploratory Analysis:** Descriptive statistics, distributions, correlation matrices.
- **Aggregations & Grouping:** Group-by event type, country, continent, season, year.
- **KPI Calculations:** Total Events, Total Deaths, Avg Severity, Avg Response Time, Aid Coverage.
- **Visualisation:** Line charts, bar charts, heat maps, pie charts, scatter plots.
- **Dashboard Preparation:** Final dataset exports and DAX/calculated field definitions.

4.2 Coding Standards

- Meaningful, descriptive variable and column names throughout.
- Consistent code formatting (PEP-8 for Python; indented CTEs for SQL).
- Inline comments explaining non-trivial logic.
- Modular structure — cleaning functions separated from analysis and visualisation.
- Reusable aggregation and KPI generation utilities.

4.3 Error Handling & Data Quality

- Missing value handling: numerical columns imputed with median; categorical with mode or "Unknown".
- Duplicate row detection and removal prior to analysis.
- Data type validation and explicit casting (e.g., date strings to DATE, numeric strings to DECIMAL).
- Post-aggregation verification: cross-checked totals across all five tools.
- Outlier identification using IQR method; flagged but retained for full analysis.

4.4 System Requirements

Component	Requirement
Python	Version 3.8 or later; libraries: pandas, numpy, matplotlib, seaborn, plotly
Jupyter Notebook	Version 6.x or later (Anaconda recommended)
Microsoft Excel	Office 2019 / Microsoft 365 (with Power Query)
Power BI	Power BI Desktop (latest version)
Tableau	Tableau Public or Tableau Desktop
SQL	MySQL 8.0 or later (or compatible RDBMS)

5 Testing & Quality Assurance

5.1 Test Cases & Test Plan

Test Case	Test Description	Expected Result	Status
TC-01	Missing value handling	Dataset free of null values after cleaning	Passed
TC-02	KPI — Total Events	COUNT(*) matches row count in cleaned dataset	Passed
TC-03	KPI — Total Deaths	SUM(deaths) consistent across SQL, Python, Power BI, Tableau	Passed
TC-04	KPI — Economic Impact	SUM(economic_impact_million_usd) matches across tools	Passed
TC-05	Dashboard filters	Country/event-type filters correctly update all visuals	Passed
TC-06	SQL aggregations	GROUP BY queries return correct grouped outputs	Passed
TC-07	Duplicate row check	No duplicate event_id values in cleaned dataset	Passed
TC-08	Data type validation	All date columns parsed as DATE; numerics as DECIMAL/INT	Passed

5.2 Bug Reports & Resolutions

Issue	Root Cause	Resolution
Incorrect data types on import	Numeric columns stored as VARCHAR in raw CSV	Applied explicit CAST() in SQL; astype() in Python
Duplicate event rows	Raw dataset contained duplicate records	Removed duplicates using DROP DUPLICATES / DISTINCT
Visualisation formatting issues	Axis labels overlapping on rotated bar charts	Adjusted font size, tick rotation, and figure dimensions
KPI inconsistency between tools	Different handling of NULL values	Standardised NULL exclusion logic across all tools

6

Key Insights & Recommendations

6.1 Key Findings

Consistently High Event Frequency: Climate events remained at persistently elevated levels throughout the 2020–2025 period, with no statistically significant downward trend, underscoring the ongoing escalation of climate risk.

Summer — Highest Severity & Casualties: The summer season recorded the highest average event severity scores and the greatest number of direct casualties, driven predominantly by heatwaves, wildfires, and tropical storms.

Autumn — Greatest Economic Loss: Autumn generated the highest aggregate economic impact across the study period, associated with the tail-end of hurricane season and elevated flood events.

Strong Seasonal Patterns by Event Type: Several event types — particularly wildfires, cold waves, and tropical cyclones — demonstrated pronounced seasonal clustering, enabling predictable peak-risk windows.

Response Time Correlates with Casualties: Events with slower emergency response times were associated with higher casualty-to-affected-population ratios, indicating that faster mobilisation meaningfully reduces human impact.

Low Aid Coverage: International aid covered approximately 0.48% of total recorded economic losses on average, highlighting a substantial funding gap relative to the scale of disaster impact.

Geographic Concentration: The United States, India, China, Japan, and Portugal recorded the highest climate-event counts; the United States alone accounted for the largest share of economic losses.

6.2 Recommendations

- Improve and invest in emergency response systems to reduce mobilisation time, particularly in high-frequency disaster regions.
- Increase national and international investment in climate-resilient infrastructure to reduce economic losses from recurring events.
- Implement evidence-based disaster preparedness strategies informed by seasonal risk calendars derived from historical data.
- Strengthen seasonal risk management plans, with special attention to summer heatwaves and autumn flooding.
- Improve international cooperation and increase aid commitments during major climate disasters to close the aid-coverage gap.
- Develop geospatially targeted early-warning systems for the highest-risk countries identified in this analysis.

6.3 Suggested Future Improvements

- Integrate machine learning models for predictive severity and casualty estimation.
- Expand geospatial analysis with choropleth maps and spatial clustering.
- Connect dashboards to live climate data APIs for near-real-time monitoring.
- Extend the dataset beyond 2025 to capture longer-term climate trends.
- Add economic recovery timelines to measure post-disaster resilience.

7

Conclusion

The **Global Climate Events Economic Impact** project successfully delivered a comprehensive, multi-tool analysis of worldwide climate disasters and their economic and human consequences over the 2020–2025 period. By applying a rigorous, standardised analytical workflow across Microsoft Excel, SQL, Python, Power BI, and Tableau, the project demonstrated both the consistency of the underlying data and the complementary strengths of each platform.

The analysis confirmed that climate events are a persistent and escalating global risk. Summer and autumn emerged as the highest-impact seasons for human casualties and economic losses, respectively, while strong seasonal patterns in specific event types provide actionable intelligence for targeted preparedness measures. The correlation between emergency response speed and casualty outcomes reinforces the critical importance of investment in rapid-response infrastructure and international coordination mechanisms.

The interactive dashboards developed in Power BI, Tableau, and Excel translate these complex findings into accessible, stakeholder-ready tools that can directly support data-driven decision making in disaster management, policy formulation, and resource allocation. The well-documented codebase and star-schema database provide a scalable foundation for future extensions — including predictive modelling, real-time data integration, and expanded geospatial analysis — as recommended in Section 6.

This project demonstrates that systematic data analysis, when applied rigorously and communicated clearly, can transform raw climate event records into meaningful insights capable of informing life-saving decisions. The team acknowledges the collaborative effort across all five members and the value of applying multiple analytical environments to validate and enrich the findings.

Appendix A — Dataset Information

Source: Kaggle — Global Climate Events Dataset

Coverage: 2020–2025 | Global scope | 51 affected countries

Total Records: Approximately 3,000 climate event records

Key Columns: Event Type, Country, Continent, Date, Season, Severity, Duration, Affected Population, Deaths, Injuries, Economic Impact (M USD), Infrastructure Damage Score, Response Time (hrs), International Aid (M USD).